

Parameter Lab Exercises

1. For the provided memory RTL, all ports and signals are hardcoded. Use parameters to make this rtl reusable with different configurations
2. Write a reg_n module with a WIDTH parameter (default 8), a synchronous reset, and an enable. Instantiate it three times in a testbench with widths 4, 8, and 16, overriding once by position and once by name (`#(.WIDTH(16))`). Can you use both at the same instance ?
3. Write a parameterized memory interface, for the signals [address, data_in, data_out, write_en] use default values for paramters, and use type parameters.
4. Make a module with parameter `int A_W = 8`, parameter `int B_W = 8` and a localparam `int OUT_W = A_W + B_W` for a multiplier's output width. Then add a second version where the default of one parameter is an expression of the other (parameter `int B_W = A_W`). Explore what happens when you override only `A_W`.